ICSE – Computer Applications (Class IX)

Semester – 1 Examination
Full Marks: 100 | Time: 2 Hours
Answer all questions. Use proper Java syntax.

Section A – MCQs (10 × 1 = 10 marks)

1. Which of the following is a valid identifier in Java?
   (a) 9name
   (b) total_marks
   (c) #value
   (d) double

2. The default value of a `boolean` variable in Java is:
   (a) 0
   (b) 1
   (c) true
   (d) false

3. The result of the expression `15 % 4` is:
   (a) 3
   (b) 4
   (c) 2
   (d) 1

4. Which of the following data types is used to store characters in Java?
   (a) int
   (b) char
   (c) String
   (d) float

5. Which method of the Math class is used to find the square root of a number?
   (a) Math.sqrt()
   (b) Math.pow()
   (c) Math.square()
   (d) Math.abs()

6. The value of the expression `(A++ + --B)` + `++C` (A= 5, B = 3, C=7)is:
   (a) 14
   (b) 13
   (c) 15
   (d) 16

7. Which of the following statements about Java operators is correct?
   (a) `++i` increases i after use
   (b) `i++` increases i before use
   (c) `++i` increases i before use
   (d) Both (a) and (c)

8. What is the range of the `byte` data type in Java?
   (a) 0 to 255
   (b) -255 to +255
   (c) -128 to +127
   (d) -32768 to +32767

9. Which keyword is used in Java to select one block of statements among multiple options?
   (a) if
   (b) switch
   (c) else
   (d) case

10. The value of `Math.pow(2, 3)` is:
    (a) 6
    (b) 8
    (c) 9
    (d) 12

Section B – Output Questions (2 × 5 = 10 marks)

Q11. Predict the output of the following code:

```
int x = 5;

System.out.println(x++ + ++x);

System.out.println(x);
```

Q12. Predict the output:

```
double a = Math.sqrt(81) + Math.pow(2, 5) - Math.abs(-10);

System.out.println(a);
```

Section C – Data Types, Operators, and Math Functions (5 × 4 = 20 marks)

Q13. State the difference between **implicit type conversion** and **explicit type casting** with one example each in Java.

Q14. Evaluate the following expression:

```
y += x++ + y++ + x++ + y; // y = 3; x = 5;
```

Find the value of y.

Q15. Differentiate between `Math.floor()` and `Math.ceil()` functions with examples.

Q16. What will be the values of `a` and `b` after the execution of the following code?

```
int a = 10, b = 5;

a = ++b + a--;

b = a-- - --b;
```

Q17. Write the Java expression for the following mathematical expression:

~~(x2+y2)z+|x-y|\frac{(x^2 + y^2)}{\sqrt{z}} + |x-y|~~

$$(x^2+y^2)z + \frac{|x-y|}{(x^2+y^2)\sqrt{z}} + |x-y|$$

Section D – Programming Questions (6 × 10 = 60 marks)

Q18. (If-Else)
Write a Java program to input the age of a person. If age is below 18, display **"Not Eligible to Vote"**, else display **"Eligible to Vote"**.

Q19. (Nested If-Else)
A school has the following grading system:

- 90 – 100 : Grade A
- 75 – 89 : Grade B
- 50 – 74 : Grade C
- Below 50 : Grade D

Write a program that inputs marks and displays the grade.

Q20. (Switch-Case Menu Driven)
Write a menu-driven program in Java to perform the following:

1. Calculate the square of a number.
2. Calculate the cube of a number.
3. Exit.

Q21. (For-Loop – Series Problem)
Write a Java program to display the series:

12+22+32+...+n

and also display the sum of the series.

Q22. (For-Loop – Number Problem)
Write a Java program to input a number and check whether it is a **prime number** or not.

Q23. (Number Problem – Logic based)
Write a Java program to input a number and check whether it is a **special number** or not.
(A special number is a number in which the sum of the factorial of its digits equals the number itself. Example: 145 → 1! + 4! + 5! = 145)
